

## Quiz Week 3



## Multiplying Permutations

---

1. Let  $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 3 & 5 \end{pmatrix}$  and  $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$ , be two permutations in  $S_5$ .

Select the correct answers:

(a)  $\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 5 & 1 & 4 \end{pmatrix}$

(b)  $\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 5 & 1 & 4 \end{pmatrix}$

(c)  $\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 2 & 5 & 3 \end{pmatrix}$

(d)  $\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 2 & 5 & 3 \end{pmatrix}$

(e)  $\tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \end{pmatrix}$

(f)  $\tau^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$

**Solution:** The correct answers are (a), (d) and (f). Computing the products  $\tau\sigma$  and  $\sigma\tau$  as explained in class, we can see that (a) and (d) are the correct ones; concerning to  $\tau^{-1}$  we can easily check that the product of  $\tau$  and the permutation given in (f) is the identity permutation, proving that (f) is correct. See all these calculations below.

$$\begin{array}{c} \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 1 & 4 & 3 & 5 \end{pmatrix} \\ \begin{array}{ccccc} & \swarrow & & \searrow & \\ & & \times & & \\ & \swarrow & & \searrow & \end{array} \\ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix} \end{array}$$

$$\tau = \left( \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 1 & 5 & 4 \end{array} \right)$$

$$\sigma = \left( \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 1 & 4 & 3 & 5 \end{array} \right)$$

$$\tau = \left( \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 1 & 5 & 4 \end{array} \right)$$

$$\left( \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 1 & 5 & 4 \end{array} \right)$$

2. Let  $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & a & b & c & d \end{pmatrix}$  and  $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ x & y & z & t & u \end{pmatrix}$  be two permutations in  $S_5$  such that  $\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 4 & 2 \end{pmatrix}$  and  $\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 & 1 \end{pmatrix}$ . Find the values of  $a, b, c, d, x, y, z, t, u$ .

**Solution:** We proceed as follows:

$$\begin{aligned} \sigma(1) &= 2, & \text{and} & & 5 = \tau\sigma(1) = \tau(2) = y &\Rightarrow y = 5 \\ \tau(2) &= 5, & \text{and} & & 4 = \sigma\tau(2) = \sigma(5) = d &\Rightarrow d = 4 \\ \sigma(5) &= 4, & \text{and} & & 2 = \tau\sigma(5) = \tau(4) = t &\Rightarrow t = 2 \\ \tau(4) &= 2, & \text{and} & & 5 = \sigma\tau(4) = \sigma(2) = a &\Rightarrow a = 5 \\ \sigma(2) &= 5, & \text{and} & & 3 = \tau\sigma(2) = \tau(5) = u &\Rightarrow u = 3 \\ \tau(5) &= 3, & \text{and} & & 1 = \sigma\tau(5) = \sigma(3) = b &\Rightarrow b = 1 \\ \sigma(3) &= 1, & \text{and} & & 1 = \tau\sigma(3) = \tau(1) = x &\Rightarrow x = 1 \end{aligned}$$

Thus  $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 1 & c & 4 \end{pmatrix}$  and  $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & z & 2 & 3 \end{pmatrix}$ . Keeping in mind that  $\sigma$  and  $\tau$  are permutations in  $S_5$  we can conclude that  $\sigma(4) = 3$  and  $\tau(3) = 4$ .

Therefore  $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 1 & 3 & 4 \end{pmatrix}$  and  $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 2 & 3 \end{pmatrix}$ .